AI-MACHINE LEARNING FOUNDATIONS

Dual Degree in Business Administration & Data and Business Analytics
BBADBA SEP-2026 AIMLF-NBDA.3.M.B

Area Computer Science
Number of sessions: 30
Academic year: 26-27
Degree course: THIRD
Number of credits: 6.0
Semester: 1º
Category: COMPULSORY
Language: English

Professor: **ADRIÁN SÁNCHEZ MORALES**

E-mail: adrians@faculty.ie.edu

Dr Sánchez holds a PhD in Artificial Intelligence from Technical University of Cartagena and Carlos III of Madrid, and a BSc in Telecommunication Engineering from the same university. He has taken part in research projects involving deep neural networks, computer vision and e-health technologies. He has several years of experience in private companies, working as a data scientist on all kind of sectors as well as managing a large amount of AI-related projects. More details in https://www.linkedin.com/in/adrián-sánchez-morales/

Office Hours

Office hours will be on request. Please contact at:

If your question cannot be properly answered via email and/or you would prefer to meet in person, please make an appointment to meet with me on the university campus during my scheduled office hours. Office hours will be determined during the semester and posted on Campus Online.

SUBJECT DESCRIPTION

Machine Learning (ML) is a core pillar of modern data analytics and artificial intelligence. Its capacity to extract patterns from data, construct predictive models, and support complex decision-making has made it an essential competency for any professional working at the intersection of data and business. From demand

forecasting and customer segmentation to fraud detection and natural language processing, ML techniques are reshaping how organizations operate, compete, and create value.

This course provides a rigorous and systematic introduction to Machine Learning, designed for students in Data and Business Analytics who seek both a solid theoretical foundation and the practical skills to build, evaluate, and critically assess ML systems. The course spans the full ML workflow: from data curation and feature engineering, through supervised and unsupervised learning, to deep learning architectures and their business applications. Reinforcement learning is introduced as a conceptual framework for sequential decision-making.

The course is structured around lectures, guided coding notebooks, two substantial take-home assignments, and a group project. Lectures develop conceptual and mathematical understanding. Notebooks demonstrate how theory translates into working code, using real datasets. Take-home assignments require students to independently build complete ML pipelines under realistic constraints. The group project requires teams to identify and address a significant, original problem using ML methods appropriate to its complexity.

Students completing this course will be equipped to work effectively in data science and analytics roles that require ML competency, to communicate ML results to technical and non-technical audiences, and to evaluate the appropriateness, quality, and limitations of ML solutions deployed in organizational contexts.

LEARNING OBJECTIVES

- Design and implement simple AI-driven applications or workflows.
- Evaluate the effectiveness of AI tools in enhancing productivity, considering factors like cost, ease of use, and performance.

By the end of this course, students will be able to:

1. Understand the landscape of ML techniques, including supervised, unsupervised, and reinforcement learning, and situate them within the broader data analytics and AI ecosystem.

2. Design and execute a complete ML project pipeline: from problem framing and data collection through feature engineering, model training, evaluation, and comparison.

3. Apply core supervised learning algorithms, including regression, classification, ensemble methods, and neural networks, selecting among them based on problem type, data characteristics, and business objectives.

4. Apply core unsupervised learning techniques, including clustering, dimensionality reduction, and anomaly detection, interpreting and communicating results in business contexts.

5. Understand the architecture and training dynamics of deep learning models, including feedforward networks, convolutional networks, recurrent networks, and Transformers, and apply them to relevant tasks.

6. Evaluate ML models rigorously using appropriate metrics, validation strategies, and comparative frameworks, and diagnose common failure modes including overfitting, data leakage, and sampling bias.

7. Implement ML pipelines in Python using industry-standard libraries, applying best practices for reproducibility, modularity, and maintainability.

8. Assess the limitations, risks, and ethical implications of ML systems deployed in business and organizational contexts, including issues of fairness, transparency, and model governance.

9. Work effectively as part of a team on a substantial, original ML project, demonstrating the capacity to scope a problem, select appropriate methods, execute a rigorous analysis, and present findings to a critical audience.

By the end of this course, students will be able to:

1. Understand the landscape of ML techniques, including supervised, unsupervised, and reinforcement learning, and situate them within the broader data analytics and AI ecosystem.

2. Design and execute a complete ML project pipeline: from problem framing and data collection through feature engineering, model training, evaluation, and comparison.

3. Apply core supervised learning algorithms, including regression, classification, ensemble methods, and neural networks, selecting among them based on problem type, data characteristics, and business objectives.

4. Apply core unsupervised learning techniques, including clustering, dimensionality reduction, and anomaly detection, interpreting and communicating results in business contexts.

5. Understand the architecture and training dynamics of deep learning models, including feedforward networks, convolutional networks, recurrent networks, and Transformers, and apply them to relevant tasks.

6. Evaluate ML models rigorously using appropriate metrics, validation strategies, and comparative frameworks, and diagnose common failure modes including overfitting, data leakage, and sampling bias.

7. Implement ML pipelines in Python using industry-standard libraries, applying best practices for reproducibility, modularity, and maintainability.

8. Assess the limitations, risks, and ethical implications of ML systems deployed in business and organizational contexts, including issues of fairness, transparency, and model governance.

9. Work effectively as part of a team on a substantial, original ML project, demonstrating the capacity to scope a problem, select appropriate methods, execute a rigorous analysis, and present findings to a critical audience.

TEACHING METHODOLOGY

IE University teaching method is defined by its collaborative, active, and applied nature. Students actively participate in the whole process to build their knowledge and sharpen their skills. Professor's main role is to lead and guide students to achieve the learning objectives of the course. This is done by engaging in a diverse range of teaching techniques and different types of learning activities such as the following:

Learning Activity	Weighting	Estimated time a student should dedicate to prepare for and participate in
Lectures	13.3 %	20.0 hours
Discussions	20.0 %	30.0 hours
Exercises in class, Asynchronous sessions, Field Work	13.3 %	20.0 hours
Group work	26.7 %	40.0 hours
Individual studying	26.7 %	40.0 hours
TOTAL	100.0 %	150.0 hours

AI POLICY

Generative artificial intelligence (GenAI) tools may be used in this course for research, code documentation, code generation, and code fixing with appropriate acknowledgement. GenAI may not be used for exams, writing conclusions, or extracting insights for the assignments. If a student is found to have used AI-generated content inappropriately, it will be considered academic misconduct, and the student might fail the respective assignment or the course.

If you are unsure whether you are using GenAI tools appropriately in this course, I encourage you to discuss your situation with me.

Below is a suggested format to acknowledge the use of generative AI tools. Please note that acknowledging AI will not impact your grade.

I acknowledge the use of [AI systems link] to [specify how you used generative AI]. The prompts used include [list of prompts]. The output of these prompts was used to [explain how you used the outputs in your work]

If AI was permitted to be used in your assignment, but you have chosen not to include any AI-rated content, the following disclosure is recommended:

No content generated by AI technologies has been used in this assignment.

PROGRAM

SESSION 1 (LIVE IN-PERSON)

Course Introduction

In this session we will review the course logistics and organization together with basic concepts and requirements along the subject.

SESSION 2 (LIVE IN-PERSON)

Introduction to Machine Learning

In this session, we will start learning about the meaning of machine learning and its objectives. We'll see some examples of learning models. Finally, we'll do some python exercises to review basic concepts of the programming language.

SESSION 3 (LIVE IN-PERSON)

Types of learning

In this session we will review the different types of learning in ML and their applications. We'll face the first practical exercise of the course.

SESSION 4 (LIVE IN-PERSON)

Supervised Learning - Regression

SESSION 5 (LIVE IN-PERSON)

Supervised Learning - Classification

In this session we will introduce some classification models such as Logistic Regression or Generative models. We will compare different classification models. Finally, we will do some related exercises.

SESSION 6 (LIVE IN-PERSON)

Overfitting and Regularization techniques

This session focuses on understanding overfitting as a key challenge in machine learning—where models capture noise and perform poorly on unseen data—and explores regularization techniques as a solution. It covers how methods like L1 and L2 introduce penalties to control model complexity, helping achieve a better balance between bias and variance and improving generalization performance.

SESSION 7 (LIVE IN-PERSON)

Practical Session - Supervised Learning

SESSION 8 (LIVE IN-PERSON)

Tree-Based methods

In this session we will see the basics of decision trees and their related ensemble methods: bagging and boosting.

SESSION 9 (LIVE IN-PERSON)

Practical Session - Trees

The idea of this session is to learn how to apply the tree-based methods to solve real problems. Essentially, we will put in practice the Random Forest and XGBoost algorithms.

SESSION 10 (LIVE IN-PERSON)

Group work introduction

SESSION 11 (LIVE IN-PERSON)

Support Vector Machines I

In these sessions we will learn the theory concepts behind the SVMs. Furthermore, we will do some python exercises about them.

SESSION 12 (LIVE IN-PERSON)

Support Vector Machines II

In these sessions we will learn the theory concepts behind the SVMs. Furthermore, we will do some python exercises about them.

SESSION 13 (LIVE IN-PERSON)

Unsupervised learning

In these sessions we will learn the challenge of unsupervised learning and the way of dealing with missing data.

SESSION 14 (LIVE IN-PERSON)

Clustering Methods

In this session we will review some of the widely known clustering methods of the literature. We will practice with some exercises.

SESSION 15 (LIVE IN-PERSON)

Multilayer Perceptron

In this session we will start learning the concepts of neural networks and deep learning. We will see the most basic network and its theory

SESSION 16 (LIVE IN-PERSON)

Convolutional Neural Networks

After knowing the basics in neural networks, we will introduce the Convolutional Neural Networks. We will review their theory and applications.

SESSION 17 (LIVE IN-PERSON)

Convolutional Neural Networks II

After knowing the basics in neural networks, we will introduce the Convolutional Neural Networks. We will review their theory and applications.

SESSION 18 (LIVE IN-PERSON)

Working with images in CV problems

Practical exercise about Computer Vision: Image classification and segmentation.

SESSION 19 (LIVE IN-PERSON)

Sequence Modeling I

After knowing the basics in neural networks, we will introduce the Recurrent Neural Networks and LSTMs. We will review their theory and applications.

SESSION 20 (LIVE IN-PERSON)

Sequence Modeling II

After knowing the basics in neural networks, we will introduce the Recurrent Neural Networks and LSTMs. We will review their theory and applications.

SESSION 21 (LIVE IN-PERSON)

Practical session - Working with text data

Practical exercise about NLP.

SESSION 22 (LIVE IN-PERSON)

Practical Session - Business AI problem

SESSION 23 (LIVE IN-PERSON)

Advanced Real Applications

In this session we will learn about transfer learning and the most common applications of the previous neural networks: computer vision and natural language processing.

SESSION 24 (LIVE IN-PERSON)

Transformers I

This session introduces **Transformers** as a foundational deep learning architecture that revolutionized AI by using self-attention mechanisms to model relationships within sequential data. It explains how transformers process entire inputs in parallel and leverage encoder–decoder structures and attention layers to capture long-range dependencies, enabling state-of-the-art performance in tasks such as language modeling, translation, and generative AI.

SESSION 25 (LIVE IN-PERSON)

Transformers II

This session introduces **Transformers** as a foundational deep learning architecture that revolutionized AI by using self-attention mechanisms to model relationships within sequential data. It explains how transformers process entire inputs in parallel and leverage encoder–decoder structures and attention layers to capture long-range dependencies, enabling state-of-the-art performance in tasks such as language modeling, translation, and generative AI.

SESSION 26 (LIVE IN-PERSON)

Retrieval Augmented Generation

This session introduces **Retrieval-Augmented Generation (RAG)** as a key approach in Generative AI that enhances large language models by combining information retrieval with text generation. It explains how RAG systems retrieve relevant data from external knowledge sources and use it to augment the model's input, enabling more accurate, up-to-date, and context-aware responses while reducing hallucinations and improving reliability.

SESSION 27 (LIVE IN-PERSON)

Group presentations I

First of three sessions dedicated to final group project presentations. Each group presents for 25 minutes followed by questions and discussion. Presentations are evaluated on problem significance, methodological rigor, quality of results and analysis, and depth of understanding demonstrated under questioning. Groups are expected to present a complete, original ML project addressing a real-world problem of non-trivial complexity.

SESSION 28 (LIVE IN-PERSON)

Group presentations II

Continuation of final group project presentations. Same format and evaluation criteria as session 27.

SESSION 29 (LIVE IN-PERSON)

Group presentations III

Continuation and conclusion of final group project presentations. Same format and evaluation criteria as sessions 27 and 28. General discussion and synthesis of themes emerging across projects.

SESSION 30 (LIVE IN-PERSON)

Quiz 2 (Final)

In-class final quiz covering all material from sessions 1 through 29. The quiz is closed-book and designed to assess both conceptual understanding and applied reasoning across the full scope of the course.

EVALUATION CRITERIA

Disclaimer: The assessment method may be subject to modifications. Should changes occur, students will be notified on the first day of class and provided with a written document detailing the modifications.

The evaluation scheme for this course reflects its pedagogical design: the majority of the grade is determined by sustained, independent work carried out over the full duration of the semester, rather than by performance in a single terminal assessment.

- **Class Participation [15%]** Active participation is expected throughout the course. Evaluation is based on the quality and not the frequency of contributions to in-class discussion. A high-quality contribution demonstrates engagement with the material, critical reasoning, and the capacity to connect concepts across sessions. Attendance is a prerequisite for participation and is tracked; students who do not meet the university's attendance requirements will be evaluated accordingly under the institutional policy. Students are encouraged to ask questions, challenge assumptions, and engage with the work of their peers. The following types of participation do not contribute positively to the evaluation: contributions that merely restate what has already been said, interventions that are off-topic or that reflect inattention, and questions that arise from failure to engage with the course material outside of class.

- **Take-home Assignments [30%]:** Several substantial take-home assignments are distributed across the semester. Each assignment has a deadline. Assignments submitted after the deadline will receive a grade of zero without exception. All assignments are individual work. A structured review session is dedicated to each assignment following the submission deadline, focusing on the most consequential errors observed across submissions. Attendance at review sessions is strongly recommended. The

use of generative AI tools is permitted for code generation, code documentation, and debugging, with mandatory acknowledgement. GenAI may not be used for the written analysis, interpretation of results, or conclusions. Failure to acknowledge the use of GenAI will be treated as academic misconduct. Use of GenAI with appropriate acknowledgement will not affect the grade.

- **Group Project and Presentation [20%]:** Groups of up to five students develop an original ML project across the second half of the semester. The project must address a problem of genuine complexity using methods appropriate to the task. Replication of existing tutorials or minor extensions of standard benchmark exercises are not acceptable. Groups are expected to make and defend non-trivial methodological choices and to produce results that reflect a serious engagement with the problem.

- **Quiz (Final) [35%]:** An in-class quiz administered at session 30, covering all material from sessions 1 through 29. The quiz is closed-book. It is designed to assess cumulative understanding of the full course, including deep learning, unsupervised learning, and reinforcement learning.

Your final grade in the course will be based on both individual and group work of different characteristics that will be weighted as follows:

criteria	percentage	Learning Objectives	Comments
Final Exam	35 %		
Group Presentation	20 %		
Individual work	30 %		
Class Participation	15 %		

RE-SIT / RE-TAKE POLICY

Each student has four chances to pass any course, distributed over two consecutive academic years: ordinary and extraordinary call (re-sit) in June/July. Students who do not comply with the 80% attendance rule during the semester will fail both calls for the current academic year and must re-enroll in the next academic year.

Students failing the course in the ordinary call will be required to sit a comprehensive re-sit examination in June/July. The re-sit examination will cover the full scope of the course. Continuous evaluation components — assignments, group project, and class participation — will not carry over to the re-sit. Students must achieve a minimum passing grade of 5 and may obtain a maximum grade of 8.0 out of 10.0 in the re-sit examination.

Re-sit examinations require physical presence at the campus of enrollment. Dates and locations will be posted in advance.

*Each student has **four** chances to pass any given course distributed over two consecutive academic years: ordinary call exams and extraordinary call exams (re-sits) in June/July.*

***Resits:** Students failing the course in the ordinary call (during the semester) will have to re-sit the exam in June/July.*

*Those students who have not complied with the **80% attendance rule** during the semester will fail both calls for this Academic Year (ordinary and extraordinary) and will have to re-take the course (i.e., re-enroll) the following Academic Year, by paying the corresponding extra cost.*

*The June/July re-sit exam will consist of a comprehensive exam. Your final grade for the course will depend on the performance in this exam only; continuous evaluation over the semester will not be taken into consideration. Students will have to achieve the minimum passing grade of 5 and can obtain a maximum grade of **8.0 (out of 10.0)** in the re-sit exam.*

***Retakers:** Students who failed the subject in a previous Academic Year and are now re-enrolled as re-takers in a course will need to check the syllabus of the assigned professor, as well as contact the professor individually, regarding the specific evaluation criteria for them as retakers in the course during that semester (ordinary call of that Academic Year). The maximum grade that may be obtained in the retake exam (3rd call) is 10.0*

*Retake students in undergraduate programs are **generally** not required to attend classes; **however, certain courses may still apply attendance requirements depending on the course methodology or specific learning objectives**. Each syllabus clearly indicates whether attendance is mandatory or not for retake students. Therefore, it's always the student's responsibility to make sure to **check the syllabus for the specific conditions for retake students** and to **contact the professor at the start of the semester** in order to find out what is required of him /her to best prepare for the retake and to pass the course. This is very important since each professor has his/her own specific requirements.*

BEHAVIOR RULES

Please, check the University's Academic Integrity Policy here. The Program Director may provide further indications.

ATTENDANCE POLICY

Please, check the University's Attendance Policy here. The Program Director may provide further indications.

ETHICAL POLICY

Please, check the University's Student Code of conduct here. The Program Director may provide further indications.